

MONCHEGORSK, Russian Federation

WMO: 222120

Lat: 67.9422N

Lon: 32.9242E

Elev: 132

StdP: 99.75

Time Zone: 3.00 (E03)

Period: 87-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	-31.0	-26.5	-34.2	0.2	-30.8	-28.8	0.3	-26.4	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	6.5	24.7	16.4	22.6	15.5	20.5	14.4	17.8	23.5	16.3	20.8	15.0	19.2	N/A	N/A

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	
15.5	11.1	19.8	14.1	10.2	18.4	13.1	9.5	17.0	50.7	24.0	46.1	21.4	42.5	19.4	21.4

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	DB	WB
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) N/A	N/A	N/A	-33.5	27.3	2.6	2.3	-35.4	28.9	-37.0	30.3	-38.4	31.6	-40.3	33.3		
(5)			-33.6	18.3	2.6	1.8	-35.5	19.6	-37.1	20.7	-38.5	21.7	-40.4	23.0		

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	1.0	-12.5	-8.8	-6.0	-0.9	5.9	10.3	14.8	12.9	8.1	1.4	-4.7	-9.1		
	DBStd	10.35	7.85	6.33	5.70	4.12	4.78	4.33	4.41	3.48	3.28	3.82	5.33	6.67		
	HDD10.0	3632	698	527	497	328	146	47	7	9	75	266	441	593		
	HDD18.3	6352	956	760	755	578	386	243	127	175	307	524	691	851		
	CDD10.0	348	0	0	0	0	19	56	157	98	18	0	0	0		
	CDD18.3	26	0	0	0	0	1	3	18	5	0	0	0	0		
	CDH23.3	122	0	0	0	0	4	16	81	21	0	0	0	0		
	CDH26.7	11	0	0	0	0	0	1	9	2	0	0	0	0		
(14) Wind	WSAvg	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A		
(15) Precipitation	PrecAvg	554	38	33	29	29	43	55	74	61	56	51	44	42		
	PrecMax	647	71	60	54	65	81	111	137	122	107	103	88	62		
	PrecMin	389	9	14	12	7	15	14	30	22	18	15	15	12		
	PrecStd	65	14	13	11	12	17	21	22	24	24	21	18	14		
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	4.5	4.1	5.7	11.4	23.0	25.7	27.9	26.0	17.5	10.6	5.2	3.4		
		MCWB	1.9	1.9	1.7	4.9	12.4	15.7	18.6	17.2	12.6	8.5	2.9	1.4		
	2%	DB	1.6	2.0	3.7	7.6	18.7	21.7	25.5	22.8	15.4	8.9	3.5	1.6		
		MCWB	0.2	0.4	0.7	3.4	10.7	13.9	16.8	16.5	11.6	6.7	2.3	0.3		
	5%	DB	-1.4	0.8	2.4	5.7	16.1	19.4	23.7	19.8	14.0	7.7	2.1	0.2		
		MCWB	-2.2	-0.3	0.2	2.6	9.1	12.4	16.1	14.8	11.2	6.0	1.4	-0.8		
	10%	DB	-3.2	-0.4	1.0	4.0	13.3	17.2	22.0	17.8	12.6	6.5	1.2	-0.9		
		MCWB	-3.9	-1.4	-0.7	1.6	8.0	11.6	15.5	14.1	10.3	5.0	0.5	-1.7		
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	2.3	2.1	2.1	5.3	13.1	17.4	19.7	19.9	13.7	9.0	3.4	1.7		
		MCDB	3.8	3.6	4.8	10.0	20.6	23.9	24.9	23.6	15.6	10.0	4.6	3.0		
	2%	WB	-0.3	0.8	1.2	3.8	11.4	14.7	18.8	16.8	12.5	7.1	2.4	0.5		
		MCDB	1.2	2.0	3.2	6.9	17.8	19.6	24.2	21.6	14.5	8.4	3.3	1.6		
	5%	WB	-2.4	-0.1	0.4	2.8	9.9	13.2	17.2	15.3	11.4	6.1	1.5	-0.6		
		MCDB	-1.6	0.8	2.1	5.1	15.5	17.4	21.9	18.6	13.6	7.4	2.2	0.2		
	10%	WB	-3.8	-1.5	-0.5	2.1	8.1	12.1	15.8	14.3	10.5	5.2	0.6	-1.8		
		MCDB	-3.2	-0.5	1.0	3.8	12.8	16.6	20.5	17.4	12.5	6.3	1.2	-0.9		
(35) Mean Daily Temperature Range	5% DB	MDBR	6.8	5.6	6.3	6.0	6.6	6.1	6.5	4.9	4.3	3.2	4.5	6.2		
		MCDBR	6.1	4.6	4.6	7.4	11.5	9.6	9.8	7.9	6.6	4.7	4.1	5.5		
		MCWBR	5.5	3.9	3.3	4.5	5.8	4.3	3.7	3.3	3.6	3.6	3.6	4.9		
	5% WB	MCDBR	6.0	5.0	4.4	6.3	10.8	7.9	8.2	6.4	6.0	3.9	4.1	5.3		
		MCWBR	5.6	4.1	3.5	3.9	6.0	4.4	3.8	3.2	3.7	3.2	3.6	4.8		
(40) Clear-Sky Solar Irradiance	taub		0.209	0.264	0.305	0.340	0.359	0.357	0.375	0.356	0.323	0.283	N/A	N/A		
	taud		1.908	2.041	2.078	2.125	2.294	2.424	2.403	2.431	2.488	2.244	N/A	N/A		
	Ebn at Noon		87	512	709	796	830	845	812	788	726	504	N/A	0		
	Edh at Noon		10	51	88	112	107	97	96	84	62	41	N/A	0		
(44) All-Sky Solar Radiation	RadAvg		0.04	0.46	1.83	3.52	4.51	4.91	4.51	3.11	1.70	0.53	0.09	0.00		
	RadStd		0.00	0.07	0.15	0.25	0.43	0.46	0.58	0.30	0.16	0.08	0.01	0.00		

Historical Trends

	DBAvg	Heating		Cooling			Degree-Days			
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
Station Only	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
Regional (12 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page